

FRAA: A Retrieval-Augmented Agent for Explainable Financial Risk Assessment

Abstract. Financial risk assessment requires calibrated predictions, timely external evidence, and explanations that can be inspected by human reviewers. Existing industrial systems often rely on manually engineered tabular features and tree ensembles, which are effective but limited in modeling long-range behavioral sequences and fast-changing market or regulatory context. This paper presents the Financial Risk Assessment Agent (FRAA), a retrieval-augmented model for explainable financial risk assessment. For each scoring request, FRAA encodes heterogeneous user behavior with a time-aware Transformer, forms a retrieval query from the current user state, retrieves timestamp-valid documents from a financial knowledge base, fuses retrieved evidence through cross-attention, and outputs both a calibrated risk score and a concise textual justification. On a proprietary 24-month multi-source dataset with over 20 million users, 1.2 billion interactions, and 15 data sources, FRAA achieves a Log Loss of 0.5163 and Recall@5 of 0.4185, reducing Log Loss by 19.6% relative to a deployed Random Forest baseline and by 9.4% relative to a non-retrieval Transformer. Ablation studies identify knowledge retrieval as the largest contributor, while retrieval-depth analysis shows that five documents give the best trade-off between evidence coverage and noise. Additional evaluations cover cross-scenario adaptation, explanation quality rated 4.32/5 by domain experts, and 15.7 ms per-query latency under the tested serving configuration.

Keywords: Financial risk assessment · Retrieval-augmented generation · Explainable AI · Risk scoring · Time-aware Transformer

1 Introduction

Financial institutions assess user-level risk from heterogeneous and time-varying evidence, including transaction histories, account states, macroeconomic indicators, regulatory disclosures, and news sentiment. Accurate risk estimates support credit review, fraud prevention, margin control, and compliance workflows [16,38,11,6,31]. In practice, many deployed systems still depend on carefully engineered tabular features and tree-based models such as Random Forests and gradient boosting [7,13,9,25]. These models remain strong baselines, but they do not naturally capture long-range sequential dependencies, and they require repeated manual updates when external conditions change.

Recent neural sequence models and Transformer architectures provide a more flexible way to model user behavior over time [17,10,39,22,36,21]. However, a

purely parametric sequence model has two limitations for financial risk assessment. First, it can only use external market or regulatory information that has already been encoded in the training data or engineered features. Second, its predictions are difficult to audit unless supported by faithful explanations. Retrieval-augmented generation (RAG) and dense retrieval methods offer a principled way to condition model decisions on external evidence [35,23,26,19], while explanation and calibration methods show the importance of making model outputs inspectable and reliable [34,29,14,4,15,32,40]. These lines of work motivate a risk model that can retrieve fresh financial evidence and explain its scoring behavior.

This work develops the **Financial Risk Assessment Agent (FRAA)**, a retrieval-augmented model for explainable financial risk assessment. FRAA is agentic in a narrow operational sense: for each scoring request, it constructs a user state, issues a retrieval query to a financial knowledge base, fuses the retrieved evidence with behavioral features, produces a risk probability, and generates a short explanation grounded in the retrieved context and influential feature groups. The term therefore refers to the retrieve-fuse-score-explain inference process, not to an open-ended autonomous agent.

The main contributions are:

- We design a retrieval-augmented financial risk model that combines time-aware sequence encoding, external evidence retrieval, cross-attention evidence fusion, calibrated risk scoring, and explanation generation.
- We evaluate FRAA on a large-scale proprietary multi-source dataset using a strict chronological split, showing consistent improvements over deployed tree-based baselines and neural sequence models.
- We report component ablations, feature-group occlusion, retrieval-depth analysis, scenario adaptation, explanation quality assessment, and latency measurements to test whether the retrieval and explanation components are necessary.

2 Related Work

2.1 Financial Risk Modeling and Tabular Baselines

Credit and financial risk prediction has long used statistical classification, scorecards, tree ensembles, support vector machines, and gradient boosting because these methods handle heterogeneous tabular features and offer strong empirical performance [16,38,3,1,28,7,13,9,25]. Prior studies also show that credit and fraud datasets are often imbalanced and operationally asymmetric, which makes ranking quality and calibration important in addition to raw classification accuracy [11,8,24,5,6,31]. In production settings, such models are often supported by feature stores and domain-specific aggregation rules. Their main weakness is not predictive strength in static tables, but the cost of maintaining hand-crafted features and the difficulty of incorporating newly emerging external evidence at inference time.

2.2 Sequential User Modeling

Sequential recommendation and behavior modeling show that recurrent networks and Transformer encoders can capture evolving user states in many settings [17,10,39,22,36]. Financial risk assessment has a similar temporal structure: recent transactions, account changes, and market signals form a non-uniform sequence whose relevance depends on timing and context. Sequence models have also been studied for credit-card fraud detection, where temporal order is central to identifying abnormal behavior [21]. FRAA follows this direction but adapts it to risk scoring, where calibration and explanation are as important as ranking performance.

2.3 Retrieval-Augmented and Explainable AI

Dense retrieval and RAG methods condition model outputs on non-parametric external evidence [35,23,26,19]. Modern text encoders such as BERT and Sentence-BERT provide practical representations for document retrieval and semantic matching [12,33]. Efficient approximate nearest-neighbor methods and vector search libraries such as Facebook AI Similarity Search (FAISS) make such retrieval practical at large scale [30,20]. In regulated domains, model decisions also need to be interpretable and calibrated. Local explanation methods such as LIME and SHAP [34,29], broader XAI surveys [14,4], and calibration analyses [15,32,40] motivate our dual-output design: FRAA reports both a risk probability and a natural-language justification grounded in retrieved evidence.

3 Method

3.1 Problem Formulation

For a user u and decision time t , the task is to estimate the probability $\hat{r}_t \in [0, 1]$ that a material adverse event occurs within the next 30 days. The label $y_t \in \{0, 1\}$ denotes whether such an event is observed. The model receives a time-ordered window of behavioral, profile, macroeconomic, and external-document signals available no later than time t . This chronological constraint is used throughout training and testing to avoid future information leakage.

Each training instance is built from an observation window ending at t and a prediction window immediately following it. Dynamic events after t are excluded, and slowly changing attributes are read from the latest snapshot available before t . External documents are handled with the same rule: a document can be retrieved only if its publication or ingestion timestamp is no later than the scoring time. This setup matches operational risk review, where decisions must be made before the adverse event and before later market information becomes available.

3.2 Multi-Source State Construction

FRAA first maps the available evidence into three feature families. The dynamic transactional context $\mathbf{d}_t \in \mathbb{R}^{D_d}$ summarizes recent transaction amount, frequency, volatility, merchant category, and event timing. The quasi-static profile context $\mathbf{s}_t \in \mathbb{R}^{D_s}$ contains slowly changing user attributes such as account tenure, product holdings, balance ranges, and risk tier. The market and macro context $\mathbf{m}_t \in \mathbb{R}^{D_m}$ contains time-aligned external indicators available before the scoring time. The external-document side of this context follows financial text studies showing that news and regulatory language can carry risk-relevant signals [37,27].

Let D_d , D_s , and D_m denote the dimensions of the three feature families. These vectors are concatenated and projected into a shared hidden space:

$$\mathbf{x}_t = \mathbf{W}_p[\mathbf{d}_t \parallel \mathbf{s}_t \parallel \mathbf{m}_t] + \mathbf{b}_p, \quad (1)$$

where $\parallel$ denotes concatenation, $\mathbf{W}_p \in \mathbb{R}^{D_h \times (D_d + D_s + D_m)}$ and $\mathbf{b}_p \in \mathbb{R}^{D_h}$ are trainable parameters, and $\mathbf{x}_t \in \mathbb{R}^{D_h}$ is the initial user-state representation.

3.3 Time-Aware Sequence Encoding

Financial behavior is irregularly spaced, so FRAA injects a time-gap encoding into each event representation. Let $\Delta\tau_i$ denote the elapsed time between consecutive events. The input to layer l of the Transformer encoder is updated as

$$\mathbf{h}_i^{(l)} = \text{TransformerBlock}^{(l)}\left(\mathbf{h}_i^{(l-1)} + \mathbf{p}_{\Delta\tau_i}\right), \quad l = 1, \dots, L, \quad (2)$$

where $\mathbf{p}_{\Delta\tau_i}$ is the time-gap embedding. The final state $\mathbf{h}_t^{(L)}$ summarizes the user’s observed behavior up to time t .

3.4 Retrieval-Augmented Evidence Fusion

To incorporate external knowledge, FRAA converts the encoded user state into a retrieval query:

$$\mathbf{q}_t = f_q(\mathbf{h}_t^{(L)}), \quad (3)$$

where $f_q(\cdot)$ is a lightweight multilayer perceptron that maps the sequence state to the document-embedding space. The financial knowledge base contains approximately 1.2 million timestamped documents, including regulatory filings, market reports, news articles, and macroeconomic commentaries. Documents are embedded offline and indexed with FAISS. At inference time, FRAA retrieves the top- K documents by cosine similarity:

$$\mathcal{D}_t = \text{topK}_{d \in \mathcal{K}_t} \frac{\mathbf{q}_t^\top \mathbf{e}_d}{\|\mathbf{q}_t\| \|\mathbf{e}_d\|}, \quad (4)$$

where $\mathcal{K}_t$ includes only documents available no later than t , and $\mathbf{e}_d$ is the embedding of document d . Document embeddings are produced by a financial-domain

text encoder initialized from bidirectional Transformer encoders and sentence-level retrieval models [12,33]. This timestamp restriction prevents future documents from leaking into the prediction.

The retrieved document embeddings are fused with the user state using cross-attention:

$$\tilde{\mathbf{h}}_t = \text{CrossAttn} \left(\mathbf{h}_t^{(L)} \mathbf{W}_Q, \mathbf{D}_t \mathbf{W}_K, \mathbf{D}_t \mathbf{W}_V \right), \quad (5)$$

where $\mathbf{D}_t \in \mathbb{R}^{K \times D_e}$ stacks the embeddings of retrieved documents, D_e is the document-embedding dimension, and $\mathbf{W}_Q$, $\mathbf{W}_K$, and $\mathbf{W}_V$ are trainable projection matrices. The fused representation $\tilde{\mathbf{h}}_t$ is passed to the scoring and explanation heads. Cross-attention is used instead of simple concatenation so that the model can assign different weights to external evidence for different user states. For example, a macroeconomic commentary may be relevant for users with concentrated exposure to a sector but weakly relevant for users whose recent transactions do not involve that sector.

3.5 Risk Scoring and Explanation Generation

The risk head outputs a calibrated probability:

$$\hat{r}_t = \sigma(\mathbf{w}_r^\top \tilde{\mathbf{h}}_t + b_r). \quad (6)$$

where $\sigma(\cdot)$ is the sigmoid function, $\mathbf{w}_r$ is a trainable risk-head vector, and b_r is a scalar bias. In parallel, the explanation head generates a concise textual justification by attending to the retrieved documents and feature-group importance signals. The training objective combines per-instance binary cross-entropy for risk prediction and a supervised explanation-fidelity term:

$$\mathcal{L} = -y_t \log \hat{r}_t - (1 - y_t) \log(1 - \hat{r}_t) + \lambda \mathcal{L}_{\text{exp}}(\hat{e}_t, e_t^{\text{ref}}), \quad (7)$$

where $\hat{e}_t$ is the generated explanation, e_t^{ref} is an expert or rule-supported reference explanation, $\mathcal{L}_{\text{exp}}$ penalizes mismatches with reference evidence and feature-group rationales, and λ balances prediction and explanation supervision. The explanation head is trained to reference retrieved evidence and influential feature groups rather than to produce a free-form rationale disconnected from the risk score. In all reported experiments, the Transformer uses 8 attention blocks, hidden dimension 256, and embedding dimension 512; retrieval depth is $K = 5$ unless otherwise stated.

Figure 1 summarizes the full inference path from multi-source evidence encoding to retrieval, fusion, scoring, and explanation generation.

4 Experiments

4.1 Dataset and Evaluation Protocol

The dataset is a proprietary 24-month multi-source corpus from a financial risk environment. It contains over 20 million users, 1.2 billion interactions, and 15

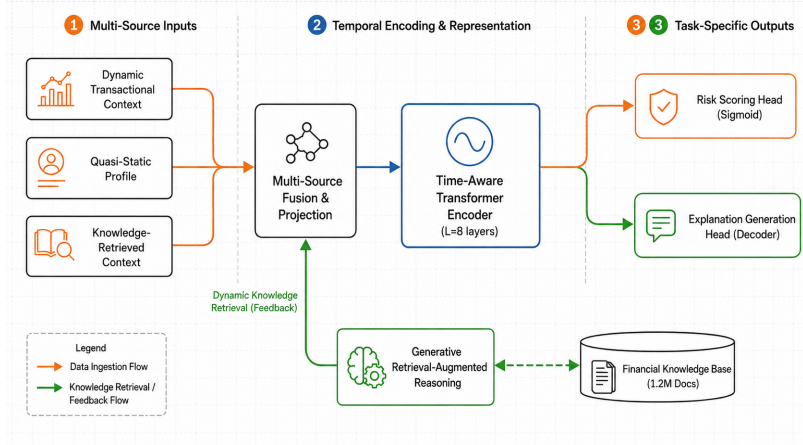


Fig. 1. Overview of FRAA. The model encodes multi-source financial behavior, retrieves timestamp-valid documents from a financial knowledge base, fuses retrieved evidence with the user state, and outputs both a calibrated risk score and an explanation.

heterogeneous data sources. Records include transaction events, account status snapshots, product holdings, macroeconomic series, and timestamped news or regulatory signals. The target is whether a material adverse event, such as default, severe delinquency, or forced liquidation, occurs within the next 30 days.

We use a strict chronological split. Historical records are used to build look-back features and timestamp-valid knowledge-base entries; the model is trained on a 12-month period, tuned on the following month, and evaluated on the final month. No validation or test-period documents are made available to earlier predictions. We report Log Loss for probability calibration, Recall@1/Recall@5 for risk-review ranking, area under the receiver operating characteristic curve (AUC) for feature-group occlusion, and longest common subsequence ROUGE (ROUGE-L) for explanation overlap with expert-written references.

The evaluation protocol is designed around the main failure modes of financial risk systems. Log Loss measures whether the predicted probabilities are useful for downstream thresholding and risk appetite control. Recall@1 and Recall@5 measure the ranking quality of the highest-risk cases, which are the cases most likely to be reviewed by analysts. The chronological split and timestamp-valid retrieval rule are used for every model variant, including ablations, so that performance differences are not caused by different access to future information.

4.2 Baselines and Training Details

We compare FRAA against a deployed Random Forest (RF), RF with static knowledge-base (KB) features, XGBoost, LSTM, and a non-retrieval Transformer. These baselines cover the main operational choices available in the target

setting: tabular ensembles, tabular ensembles with manually attached external features, recurrent sequence modeling, and neural sequence modeling without retrieval. We keep the deployed tree-based baseline because prior credit-scoring and tabular-learning studies repeatedly find tree ensembles difficult to displace on heterogeneous tabular data [3,8,2]. All baselines use the same chronological split and available feature families. FRAA is implemented in PyTorch and trained with AdamW, learning rate 1×10^{-4} , weight decay 1×10^{-5} , batch size 64, and a maximum budget of 50 epochs. The selected checkpoint is the one with the lowest validation Log Loss.

Hyperparameters are chosen on the validation month and kept fixed for the test month. For the retrieval variants, document embeddings are constructed offline and indexed before evaluation. During scoring, the retrieval pool is filtered by timestamp before nearest-neighbor search. This procedure is important because a high-capacity retrieval model can otherwise learn from documents that were not available when the decision should have been made.

4.3 Evaluation Questions

The experiments are organized around four questions. First, does retrieval-augmented evidence improve calibrated risk prediction and high-risk ranking compared with tree-based and neural baselines? Second, do the learned representations transfer to related financial scenarios, or are they tied to a single product-specific task? Third, which components of FRAA account for the improvement, especially the dynamic transaction encoder, time-gap encoding, and knowledge retrieval module? Fourth, can the model provide explanations and latency that are compatible with analyst-facing risk review? The tables in the following sections follow this order so that the main predictive result is separated from auxiliary analyses.

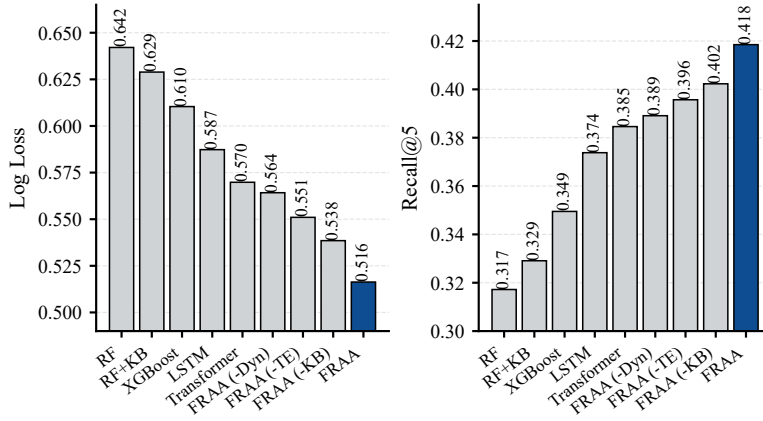
4.4 Main Results

This experiment tests the first question by comparing the full model with deployed and neural baselines under the same chronological split. Table 1 reports the numerical results, while Fig. 2 visualizes the same comparison for the two primary metrics. The ordering is consistent across calibration and ranking metrics: the full retrieval-augmented model outperforms tree-based baselines, the non-retrieval Transformer, and all ablated FRAA variants.

Relative to the deployed RF baseline, FRAA reduces Log Loss by 19.6% and improves Recall@5 from 0.3172 to 0.4185. Relative to the non-retrieval Transformer, FRAA reduces Log Loss by 9.4% and improves Recall@5 by 8.8%. The ablated variants remain below the full model, indicating that the retrieval, temporal, and dynamic-feature components contribute complementary information.

Table 1. Main test-set performance. Lower Log Loss and higher Recall@5 are better.

Model	Log Loss	Recall@5
RF (deployed baseline)	0.6421	0.3172
RF + KB features	0.6289	0.3291
XGBoost	0.6104	0.3495
LSTM	0.5873	0.3738
Transformer	0.5698	0.3846
FRAA (-Dyn)	0.5642	0.3891
FRAA (-TE)	0.5510	0.3957
FRAA (-KB)	0.5385	0.4023
FRAA (full)	0.5163	0.4185

**Fig. 2.** Main-result comparison. FRAA obtains the lowest Log Loss and highest Recall@5 among all evaluated models.

4.5 Scenario Adaptation

This experiment evaluates whether the representation learned from the multi-source corpus can be adapted to related risk scenarios with different product distributions.

Table 2 evaluates whether representations learned from the multi-source risk corpus transfer to related financial scenarios. Linear-probe fine-tuning (LP-FT) updates only a small prediction layer, low-rank adaptation fine-tuning (LoRA-FT) updates low-rank adapter parameters [18], and full fine-tuning (full-FT) updates the full model. Full-FT produces the largest gains, while LoRA-FT retains most of the benefit with fewer updated parameters. These results support the claim that FRAA learns reusable cross-product risk representations rather than only fitting a single product-specific task.

Table 2. Cross-scenario adaptation measured by relative Recall lift over scenario-specific baselines.

Adaptation method	Credit card R@1 / R@5	Mortgage R@1 / R@5
Scenario-specific model	0.00 / 0.00	0.00 / 0.00
LP-FT (< 1% params)	+3.14 / +2.97	+2.88 / +2.45
LoRA-FT (< 5% params)	+13.49 / +12.63	+11.72 / +10.88
FRAA full-FT	+18.21 / +17.10	+16.45 / +15.33

5 Ablation and Analysis

5.1 Architectural Components

The component ablation removes one design choice at a time while keeping the training split, feature availability, and evaluation metrics unchanged. The goal is to distinguish improvements due to retrieval from improvements due to stronger sequence modeling.

Table 3. Component ablation on the risk-score prediction task.

Model variant	Log Loss	Recall@5
Transformer	0.5698	0.3846
FRAA (-Dyn)	0.5642	0.3891
FRAA (-TE)	0.5510	0.3957
FRAA (-KB)	0.5385	0.4023
FRAA (full)	0.5163	0.4185

Table 3 isolates the contribution of dynamic transactions (-Dyn), temporal encoding (-TE), and knowledge retrieval (-KB). Removing the knowledge component causes the largest degradation among FRAA variants: Log Loss increases from 0.5163 to 0.5385 and Recall@5 decreases from 0.4185 to 0.4023. This supports the retrieval-augmented design. Removing temporal encoding also hurts performance, confirming the importance of irregular event timing in risk assessment. The -Dyn variant remains stronger than the non-retrieval Transformer, which suggests that external evidence and temporal encoding can partially compensate for a weaker dynamic-feature branch, but they do not replace it.

5.2 Feature Groups and Retrieval Depth

We next examine the source of the gain from two complementary views: feature-family occlusion and the number of retrieved documents.

The occlusion results in Table 4 are consistent with the component ablation. Knowledge-retrieved context produces the largest AUC drop, followed by

Table 4. Feature-group occlusion. A larger AUC drop indicates a more important feature family.

Occluded feature group	AUC drop (%)
Knowledge-retrieved context	3.92
Transaction dynamics	2.87
Macroeconomic indicators	1.55
Quasi-static profile	0.93

transaction dynamics. Static profile variables still contribute, but they are less important than time-sensitive behavioral and external-evidence signals. This ordering is useful operationally: it indicates that most of the model’s advantage comes from information that changes over time rather than from static attributes alone.

Table 5. Effect of retrieval depth K .

Top- K documents	Log Loss	Recall@5
1	0.5287	0.4068
3	0.5214	0.4117
5	0.5163	0.4185
7	0.5175	0.4173
10	0.5189	0.4160

Table 5 indicates that retrieval depth has a clear optimum. Retrieving only one or three documents provides insufficient external context, while retrieving seven or ten documents slightly degrades both metrics. We therefore use $K = 5$ in the main experiments. The degradation at larger K is small but consistent with the expected retrieval-noise trade-off: broader retrieval increases coverage, but it also introduces less relevant documents into the fusion module.

6 Further Evaluation

6.1 Explanation Quality

Beyond risk scores, FRAA is intended to support human review by exposing why a high-risk case was flagged. We therefore evaluate both expert ratings and lexical overlap with reference explanations.

Five domain experts evaluated 1,000 high-risk test predictions. Table 6 reports higher faithfulness, readability, and ROUGE-L for FRAA than for a template baseline that lists the top three feature groups. This supports the value of the explanation head beyond feature enumeration. The expert ratings should be

Table 6. Human and automatic evaluation of generated explanations. Higher scores are better.

Method	Faithfulness	Readability	ROUGE-L
Template-based top-3 groups	3.89	4.02	0.3026
FRAA (full)	4.32	4.51	0.4131

interpreted as an auditability signal rather than as evidence that the generated text is sufficient for automatic decision making.

6.2 Latency and Offline Business Utility

The final evaluation checks whether the added retrieval and fusion steps are plausible in an analyst-facing review loop and whether the improved calibration translates into offline decision-support utility.

Table 7. Inference latency and throughput under the tested batch-size-64 serving configuration.

Model	Latency (ms/query)	Throughput (queries/s)
RF + KB features	1.9	12,530
XGBoost	2.1	11,200
LSTM	8.3	5,910
Transformer	12.4	4,320
FRAA (full)	15.7	4,070

FRAA is slower than tree ensembles because it performs retrieval and neural sequence encoding, but Table 7 places its tested latency within an interactive risk-review setting. The retrieval step accounts for approximately 4.2 ms of the total latency. The throughput is lower than the tabular baselines but remains above the level required for batched offline scoring and analyst triage in the tested configuration.

Table 8. Offline replay analysis of risk-adjusted revenue (RAR) utility.

Model	Log Loss reduction (%)	RAR lift (%)
RF + KB features	-2.06	+1.89
Transformer	-11.26	+5.34
FRAA (-KB)	-16.13	+7.80
FRAA (full)	-19.59	+10.45

The offline replay in Table 8 connects calibration improvement with downstream utility under a fixed risk-appetite constraint. We report this as offline decision-support evidence, not as a claim of causal online revenue impact. The main value of this analysis is that it checks whether lower Log Loss corresponds to a practically meaningful ordering of cases under a fixed review budget.

7 Limitations and Reproducibility

The evaluation uses a proprietary financial dataset that cannot be publicly released because it contains sensitive user-level risk and transaction information. This is a limitation for independent replication. To support reproducibility within this constraint, we report the data scale, feature families, chronological split, training configuration, retrieval depth, baselines, and ablation settings. The knowledge base is timestamped, and retrieval only uses documents available at or before the scoring time. These details are intended to make the experimental protocol auditable even when the raw records cannot be shared.

Several claims should be read with this scope in mind. First, the results establish performance on one large multi-source financial environment, but not universal transfer across institutions. Second, the business-utility analysis is based on offline replay and should not be interpreted as causal online impact evidence. Third, the current evaluation does not report multi-seed variance or formal statistical significance tests, because the available audit record contains one fixed proprietary chronological split rather than repeated independent training runs. Fourth, the explanation evaluation measures expert-perceived faithfulness and readability for high-risk test predictions; it does not remove the need for human oversight in regulated decisions. Future work should add public benchmark evaluations where legally possible, repeatability analysis under multiple splits or random seeds, institution-level transfer tests, and governance mechanisms for retrieved evidence that may become outdated or disputed.

8 Conclusion

This paper presented FRAA, a retrieval-augmented model for explainable financial risk assessment. FRAA combines time-aware sequence modeling, timestamp-valid knowledge retrieval, cross-attention evidence fusion, calibrated risk scoring, and explanation generation. On a large-scale proprietary multi-source dataset, FRAA achieves the best Log Loss and Recall@5 among all evaluated models. Ablations identify knowledge retrieval as the most important component, and retrieval-depth analysis selects five documents as the best trade-off between evidence coverage and noise. Additional evaluations cover cross-scenario transfer, explanation quality, and tested latency. Overall, the results indicate that external evidence retrieval is most useful when it is constrained by time validity, fused with temporal user behavior, and exposed through auditable explanations.

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